# I Diagrammi di Nyquist - SISO

- I diagrammi sono graduati in funzione della pulsazione *w*
- I diagrammi polari sono molto importanti per lo studio della stabilità dei sistemi in retroazione (criterio di stabilità di Nyquist)
- Se conosciamo la funzione di trasferimento *G(s)*, il diagramma polare si può tracciare per punti separando le parti reali e immaginarie di *G(jw)* e determinandone i valori corrispondenti a vari valori di *w*

Partiamo dalla corrispondente funzione di risposta armonica di G(s)

$$G(j\omega) = K_1 \frac{(j\omega - z_1)(j\omega - z_2)\dots(j\omega - z_m)}{(j\omega - p_1)(j\omega - p_2)\dots(j\omega - p_n)}$$

# I Diagrammi di Nyquist - tracciare solo qualche punto

$$G(s) = k_1 \cdot \frac{b_0 + b_1 s + b_2 s^2 + \cdots + s^m}{s^h \cdot (a_h + a_{h-1} s + a_{h-2} s^2 + \cdots + s^{n-h})}$$

Supponiamo che il sistema sia di tipo 0 (h=0), non ci sono poli nell'origine

$$G(0) = \lim_{\omega \to 0} G(j\omega) = k_1 \cdot \frac{b_0}{a_0} \qquad \text{con fase: } \begin{cases} \varphi_{\omega \to 0} = 0 & \text{se } k_1 \cdot \frac{b_0}{a_0} > 0 \\ \varphi_{\omega \to 0} = -\pi & \text{se } k_1 \cdot \frac{b_0}{a_0} < 0 \end{cases}$$

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Sistema tipo 0 (h=0), non ci sono poli nell'origine

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Sistema NON tipo 0 (h=0), ci sono poli nell'origine

$$\lim_{\omega \to 0} |G(j\omega)| = \infty, \quad \text{con fase} \quad \angle G(j\omega) = \frac{1}{s^h} \quad \boxed{= -h \cdot \frac{\pi}{2}}$$

Se $K_1 > 0$

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$$G(s) = k_1 \cdot \frac{b_0 + b_1 s + b_2 s^2 + \cdots + s^m}{s^h \cdot (a_h + a_{h-1} s + a_{h-2} s^2 + \cdots + s^{n-h})}$$

$\omega \to \infty$

$$G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = \begin{cases} 0 & \text{se } n > m \\ k_1 & \text{se } n = m \end{cases}$$

$n > m$ → $G(s) \approx \frac{s^m}{s^n} = \frac{1}{s^{n-m}}$.

$n = m$ → $G(s) \approx \frac{k_1 s^m}{s^m} = k_1$. → e la fase di $K_1$

Se n>m

con fase corrispondente alla fase di

$$\frac{1}{s^{n-m}},$$

se $k_1$ ha segno positivo.

# I Diagrammi di Nyquist - tracciare solo qualche punto

$$G(s) = k_1 \cdot \frac{b_0 + b_1 s + b_2 s^2 + \cdots + s^m}{s^h \cdot (a_h + a_{h-1} s + a_{h-2} s^2 + \cdots + s^{n-h})}$$

Abbiamo tracciato il valore di *G(jw)* in due punti: w=0 e Infinito

Ed il resto del diagramma?

Per esempio, possiamo tracciare il valore di G per le intersezioni con gli assi

Usiamo il metodo analitico,

- poniamo la parte reale = 0 e troviamo l'intersezione con l'asse immaginario
- poniamo la parte immaginaria = 0 e troviamo l'intersezione con l'asse reale

# I Diagrammi di Nyquist - tracciare solo qualche punto

Esempio

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

$\frac{1}{(j\omega)^2+3(j\omega)+2}|_{\omega=0} = \frac{1}{2}, \angle = 0$

$\frac{1}{(j\omega)^2+3(j\omega)+2}|_{\omega=\infty} = \frac{1}{\infty} = 0, \angle = -180$

Convertiamo *G(jw) -> x+jy* $\frac{1}{(j\omega)^2 + 3(j\omega) + 2} = \frac{1}{(2-\omega^2) + 3\omega j} \frac{(2-\omega^2) - 3\omega j}{(2-\omega^2) - 3\omega j} = \frac{(2-\omega^2) - 3\omega j}{(2-\omega^2)^2 + 9\omega^2}$

# I Diagrammi di Nyquist - tracciare solo qualche punto

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- poniamo la parte reale = 0 e troviamo l'intersezione con l'asse immaginario

$$\frac{(2-\omega^2)}{(2-\omega^2)^2 + 9\omega^2} = 0 \Rightarrow \omega = \sqrt{2} \quad \Rightarrow \quad \frac{-3\sqrt{2}}{2 \cdot 9} = -\frac{\sqrt{2}}{6} \approx -0.236$$

# I Diagrammi di Nyquist - tracciare solo qualche punto

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- poniamo la parte immaginaria = 0 e troviamo l'intersezione con l'asse reale

$$\frac{-3\omega j}{(2-\omega^2)^2 + 9\omega^2} = 0 \Rightarrow \omega = 0 \, , \omega = \infty$$

# I Diagrammi di Nyquist - tracciare solo qualche punto

Esempio

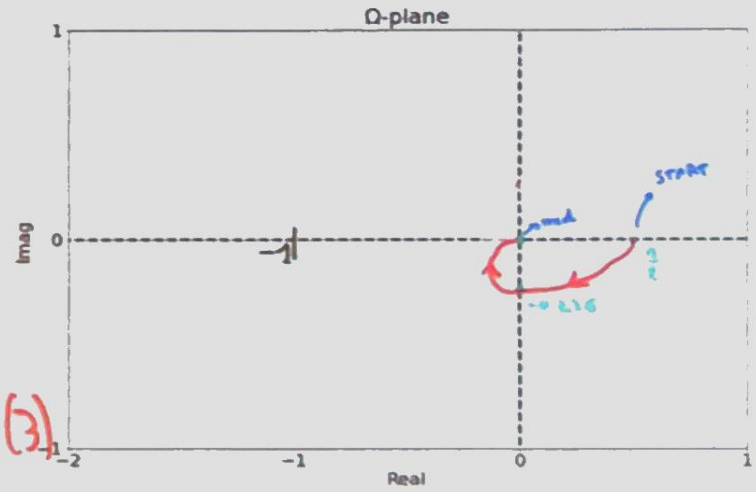
$$G(s) = \frac{1}{s^2 + 3s + 2}$$

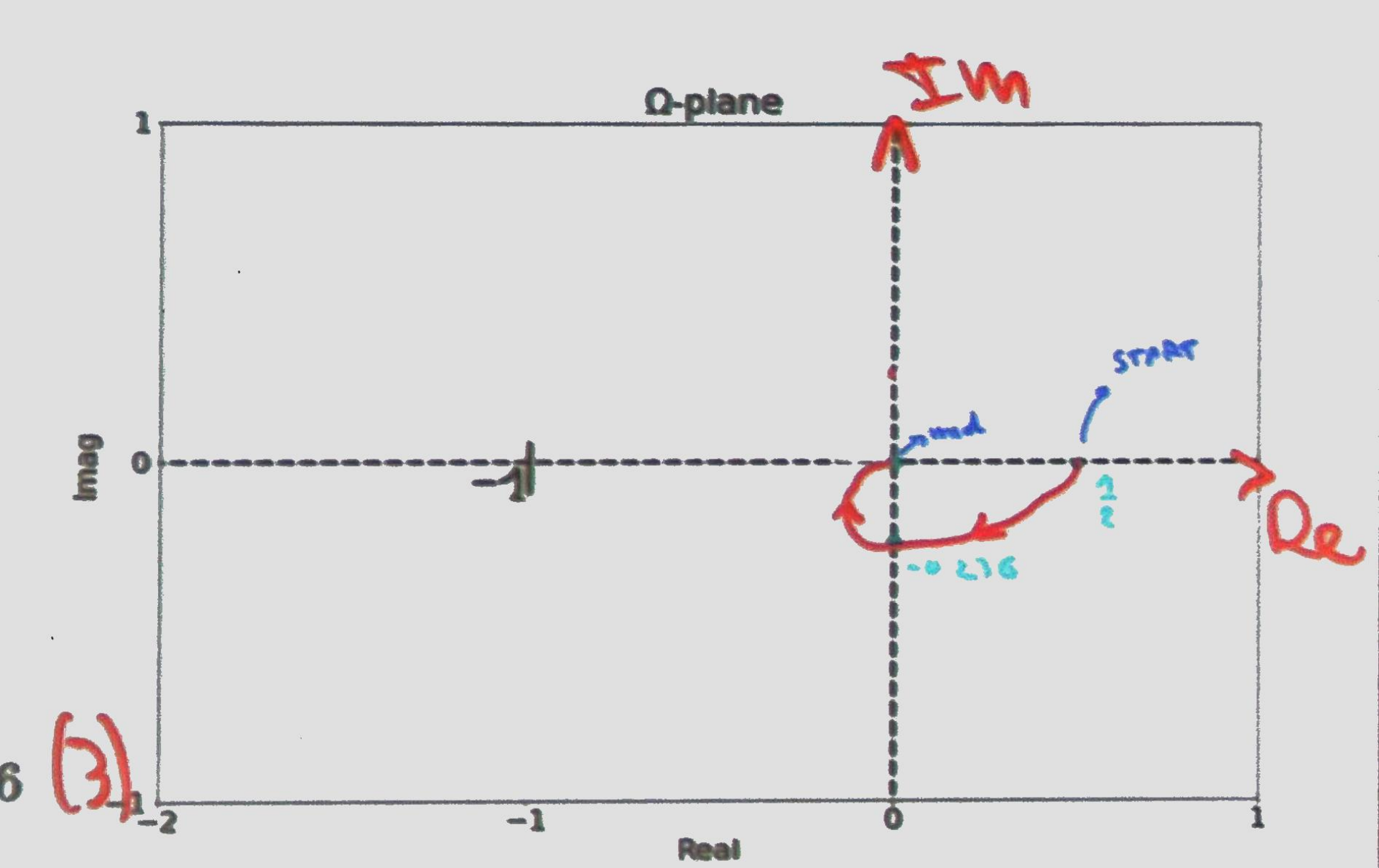
$\frac{1}{(j\omega)^2+3(j\omega)+2}|_{\omega=0} = \frac{1}{2}, \angle = 0$ (1)

$\frac{1}{(j\omega)^2+3(j\omega)+2}|_{\omega=\infty} = \frac{1}{\infty} = 0, \angle = -180$ (2)

$\frac{(2-\omega^2)}{(2-\omega^2)^2+9\omega^2} = 0 \Rightarrow \omega = \sqrt{2} \Rightarrow \frac{-3\sqrt{2}}{2 \cdot 9} = -\frac{\sqrt{2}}{6} \approx -0.236$ (3)

$\frac{-3\omega j}{(2-\omega^2)^2+9\omega^2} = 0 \Rightarrow \omega = 0\,, \omega = \infty$ (4)






6 (3)

# I Diagrammi di Nyquist - tracciare con Bode

Regola pratica:

Il diagramma di Nyquist può essere ricavato in via qualitativa a partire dalla conoscenza dei corrispondenti diagrammi di Bode, in modulo e fase

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{K}{(1 + sT_1)(1 + sT_2)(1 + sT_3)}$$

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{K}{(1+sT_1)(1+sT_2)(1+sT_3)}$$

$$G(s)|_{s=j\omega} = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)(1+j\omega T_3)}$$

per $\omega \to 0$ : $\quad G(0) = \lim_{\omega \to 0} G(j\omega) = K$ con fase 0, perché $K > 0$

# I Diagrammi di Nyquist - tracciare con Bode

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$$G(s)|_{s=j\omega} = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)(1+j\omega T_3)}$$

$\frac{k}{j\omega \cdot j\omega\, j\omega}$ polo TRIPLO

per $\omega \to 0$ : $G(0) = \lim_{\omega \to 0} G(j\omega) = K$ con fase 0, perché $K > 0$

per $\omega \to \infty$ : $G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = 0$, con la fase del polo triplo, $-270°$

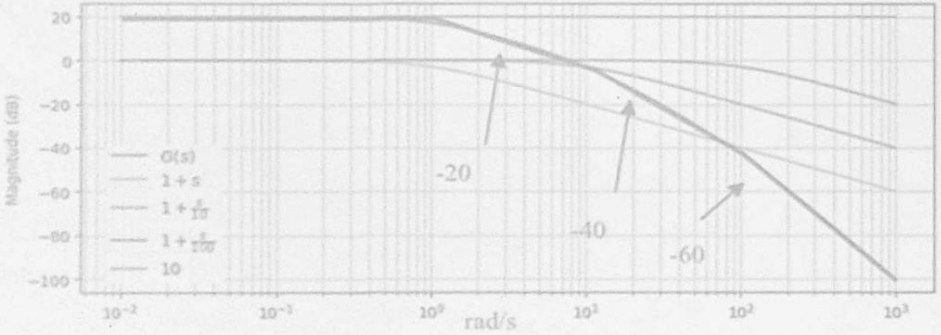
# I Diagrammi di Nyquist - tracciare con Bode

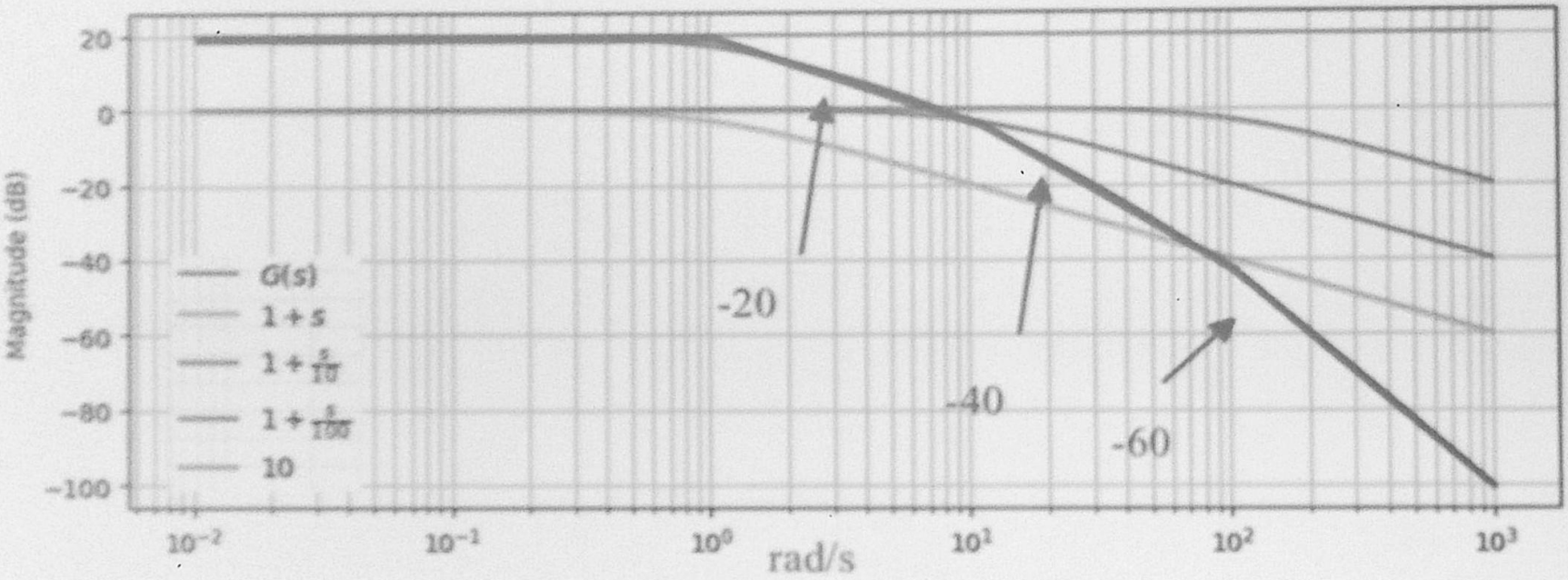
$$G(s) = \frac{K}{(1+sT_1)(1+sT_2)(1+sT_3)}$$

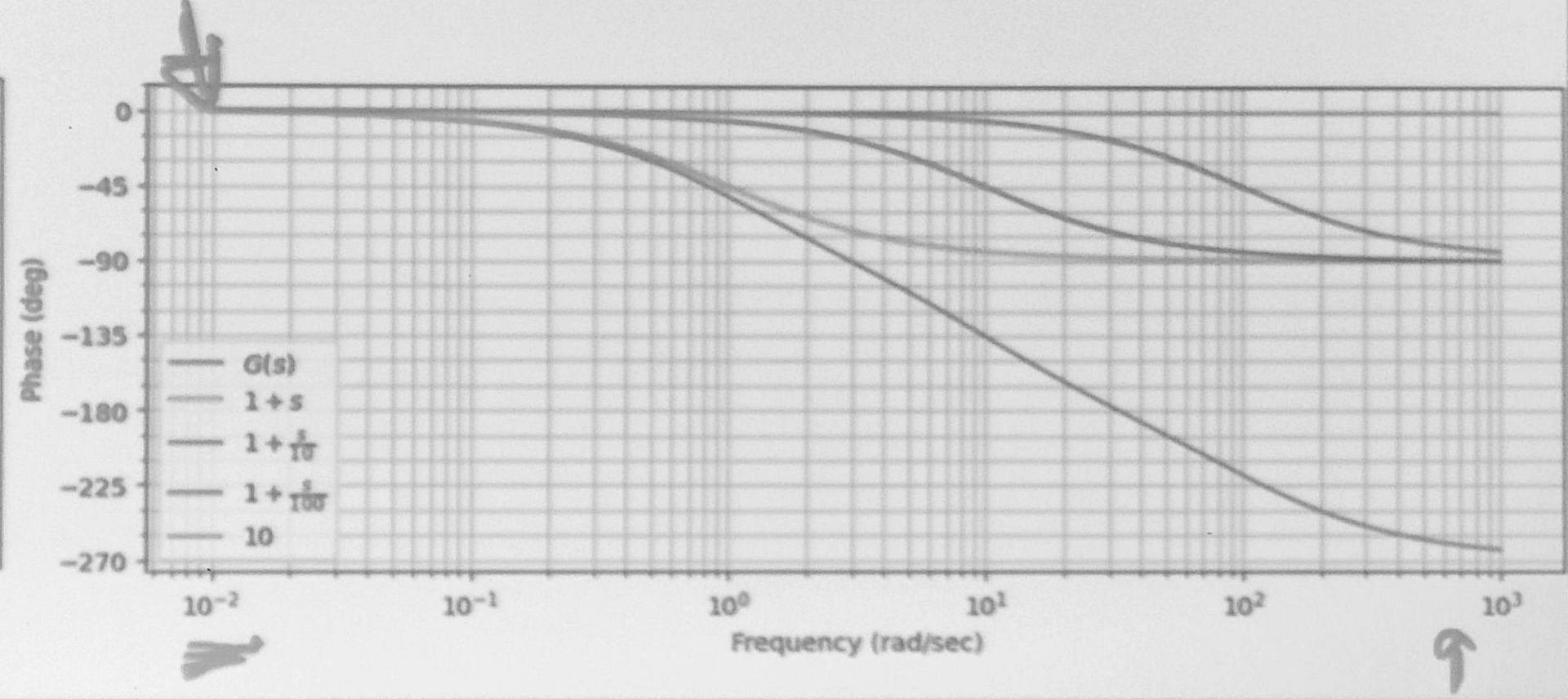
$$\boxed{G(s) = \frac{10}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10}\right)(1+s)}}$$

per $\omega \to 0$ : $\quad G(0) = \lim_{\omega \to 0} G(j\omega) = K$ con fase 0, perché $K > 0$

per $\omega \to \infty$ : $\quad G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = 0$, con la fase del polo triplo, $-270°$




20
0
−20
−40
−60
−80
−100
Magnitude (dB)
G(s)
1 + s
1 + s/10
1 + s/100
10
-20
-40
-60
10^−2
10^−1
10^0
10^1
10^2
10^3
rad/s

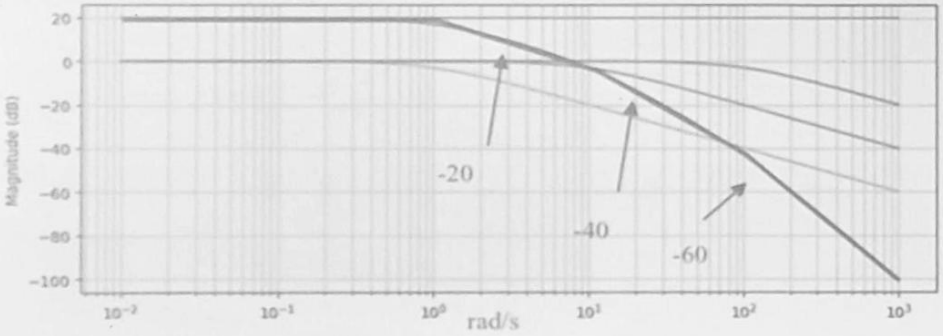

0
−45
−90
−135
−180
−225
−270
Phase (deg)
G(s)
1 + s
1 + s/10
1 + s/100
10
10^−2
10^−1
10^0
10^1
10^2
10^3
Frequency (rad/sec)

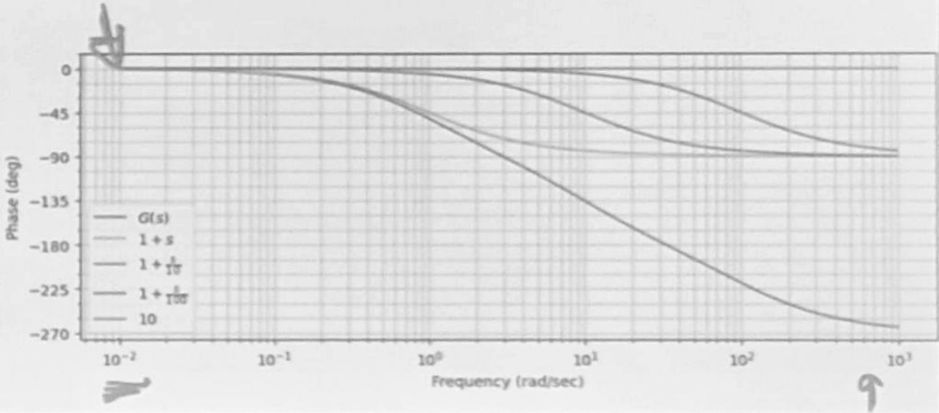
# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{K}{(1+sT_1)(1+sT_2)(1+sT_3)} \qquad G(s) = \frac{10}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10}\right)(1+s)}$$

per $\omega \to 0$ : $G(0) = \lim_{\omega \to 0} G(j\omega) = K$ con fase 0, perché $K > 0$ ✓

per $\omega \to \infty$ : $G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = 0$, con la fase del polo triplo, $-270°$ ✓



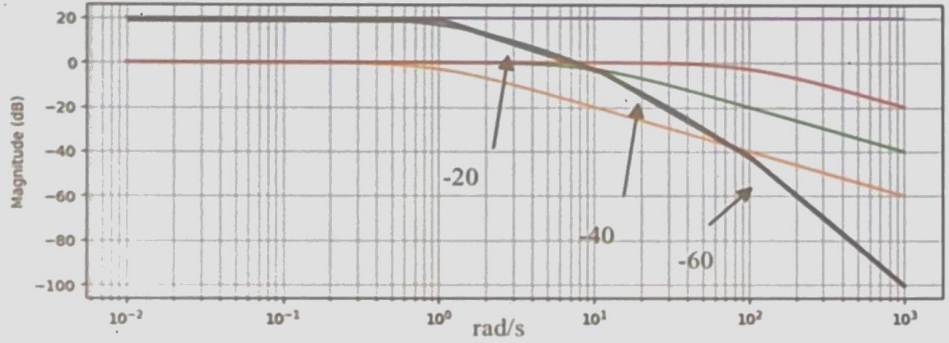



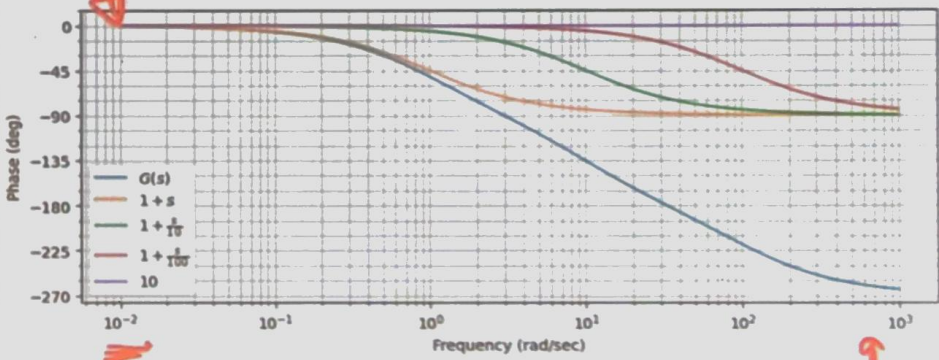
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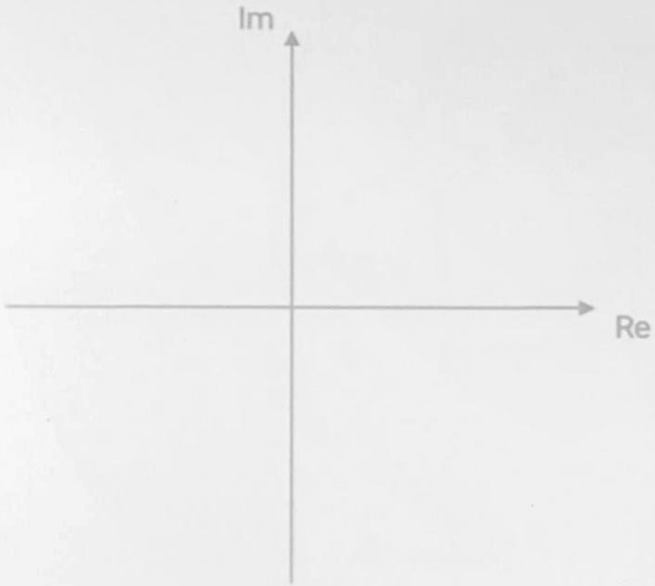
per $\omega \to 0$ : $G(0) = \lim_{\omega \to 0} G(j\omega) = K$ con fase 0, perché $K > 0$

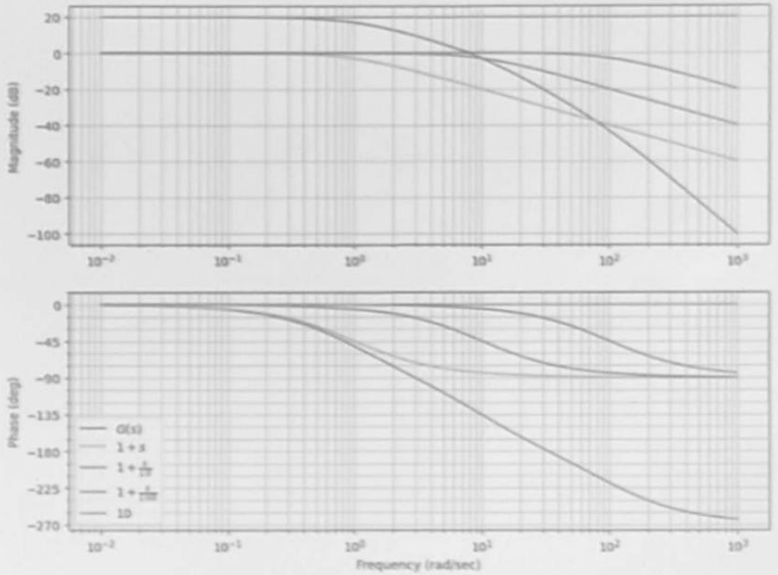
per $\omega \to \infty$ : $G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = 0$, con la fase del polo triplo, $-270°$

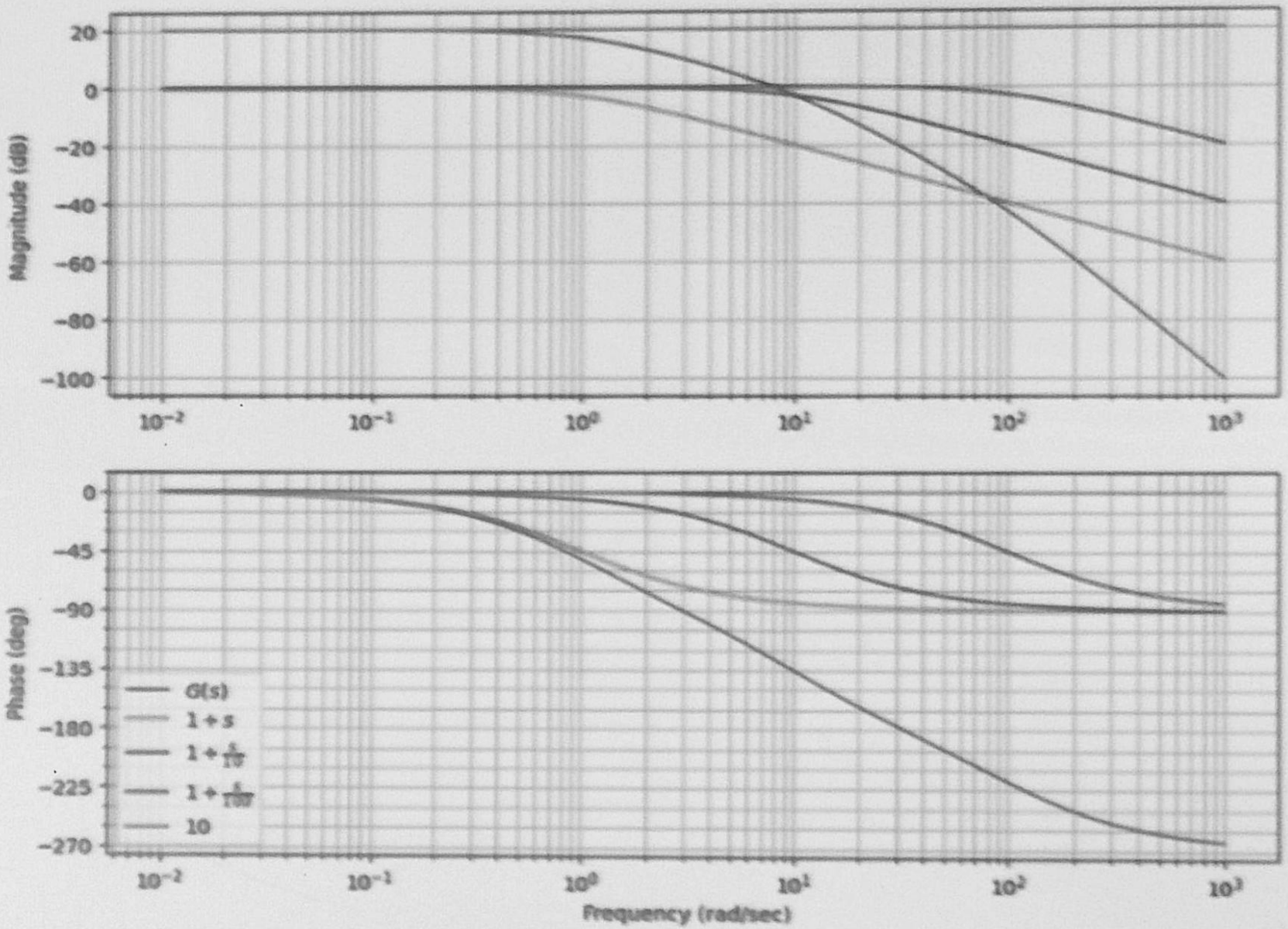


# I Diagrammi di Nyquist - tracciare con Bode




Magnitude (dB)
20
0
−20
−40
−60
−80
−100
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
$10^{3}$
Phase (deg)
0
−45
−90
−135
−180
−225
−270
G(s)
1 + s
$1 + \frac{s}{10}$
$1 + \frac{s}{100}$
10
Frequency (rad/sec)

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{K}{s(1+sT)} \longrightarrow G(j\omega) = \frac{K}{j\omega(1+j\omega T)}$$

$$G(\infty) = \lim_{\omega \to +\infty} G(j\omega) = 0$$ Con fase -180

$$G(0) = \lim_{\omega \to 0} G(j\omega) = +\infty$$ Con fase -90

Questo ci dice che c'e' un asintoto per *w=0*

- Modulo sempre decrescente
- Fase decresce da -90 a -180

# I Diagrammi di Nyquist - tracciare con Bode

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Calcolo dell'asintoto

$\mathrm{Re}(G(j\omega)) = \frac{-KT}{1+\omega^2T^2}$ $\mathrm{Im}(G(j\omega)) = \frac{-K}{\omega(1+\omega^2T^2)}$

$\lim_{\omega \to 0} Re(G(j\omega)) = -KT$

Valore finito

infinito

# I Diagrammi di Nyquist - tracciare con Bode

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$\lim_{\omega \to 0} Re(G(j\omega)) = -KT$ infinito

Valore finito

Metodo alternativo

$$\frac{1}{1+x} \approx 1 - x$$

Con Taylor, per $x \to 0$

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{K}{s(1+sT)} \longrightarrow G(j\omega) = \frac{K}{j\omega(1+j\omega T)}$$

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Calcolo dell'asintoto

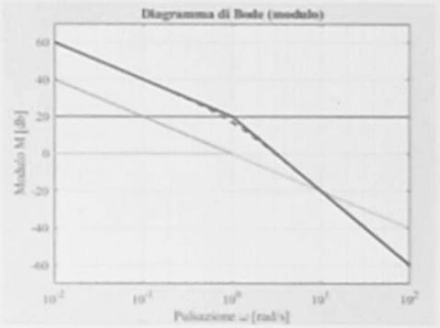
$\angle G(j\omega)$

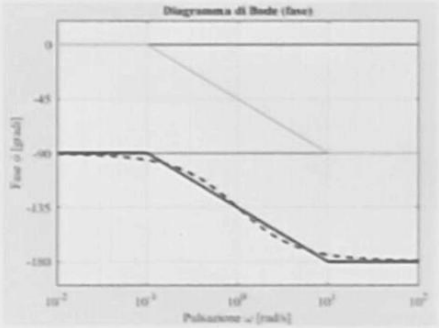
$$\frac{1}{1+x} \approx 1 - x \longrightarrow \frac{K}{j\omega(1+j\omega T)} \approx \frac{K}{j\omega}(1 - j\omega T), \quad \omega \to 0$$

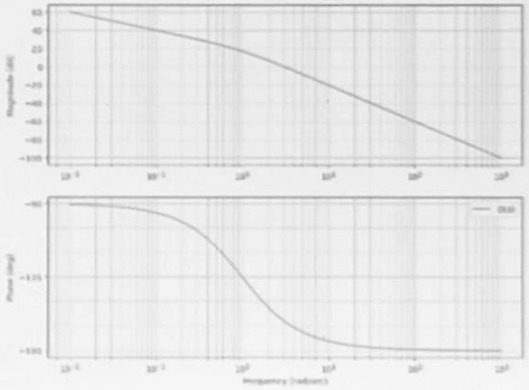
Vedo subito che la parte reale e' -KT

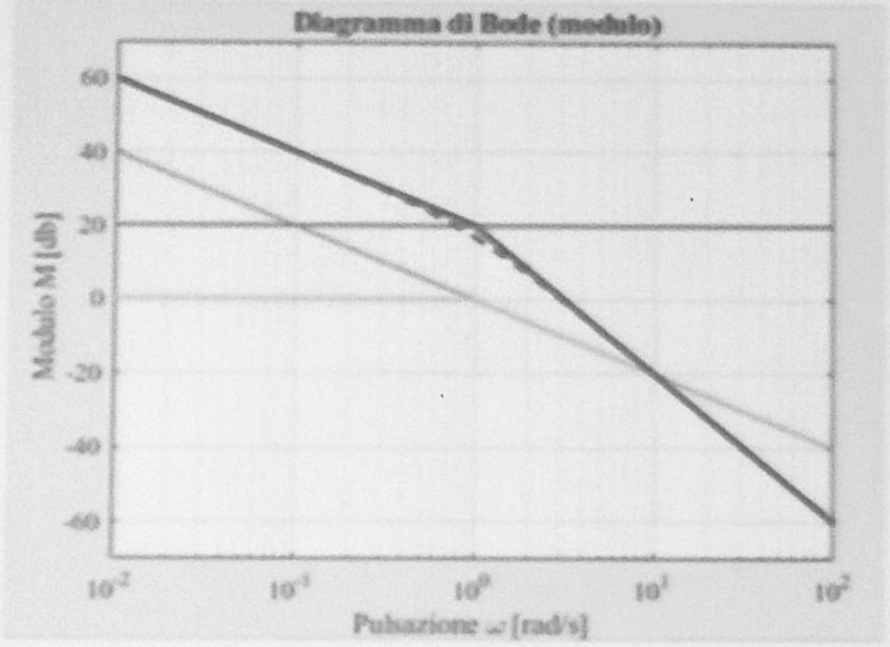
# I Diagrammi di Nyquist - tracciare con Bode

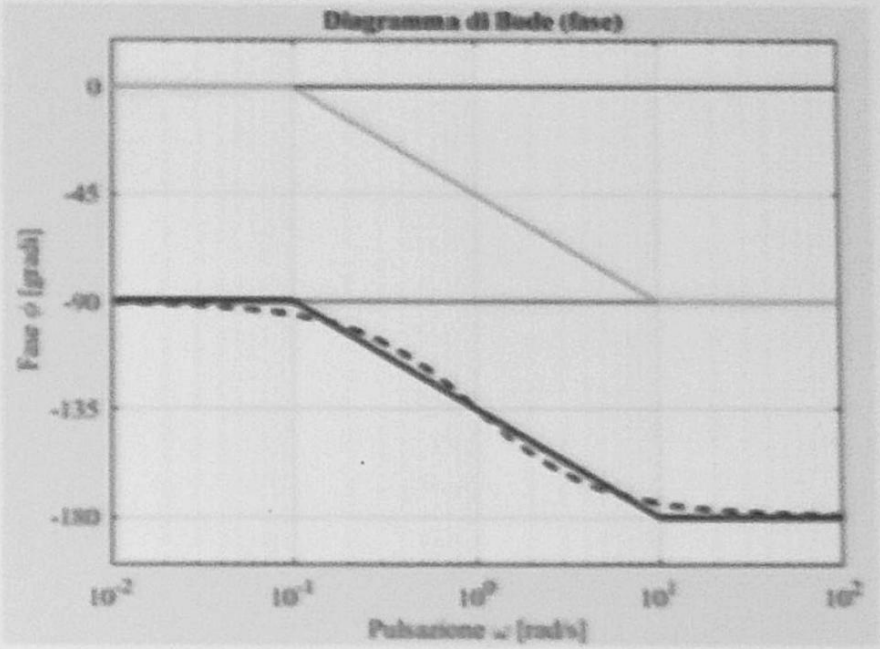
$$G(s) = \frac{10}{s(1+s)}$$

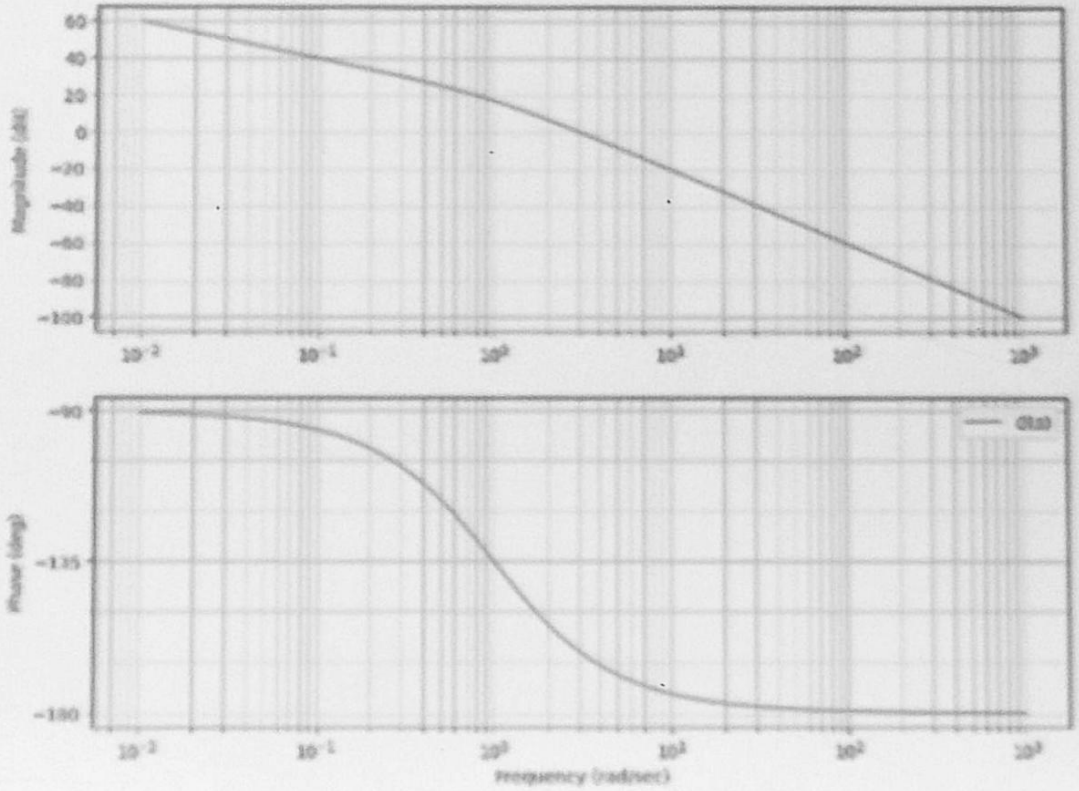


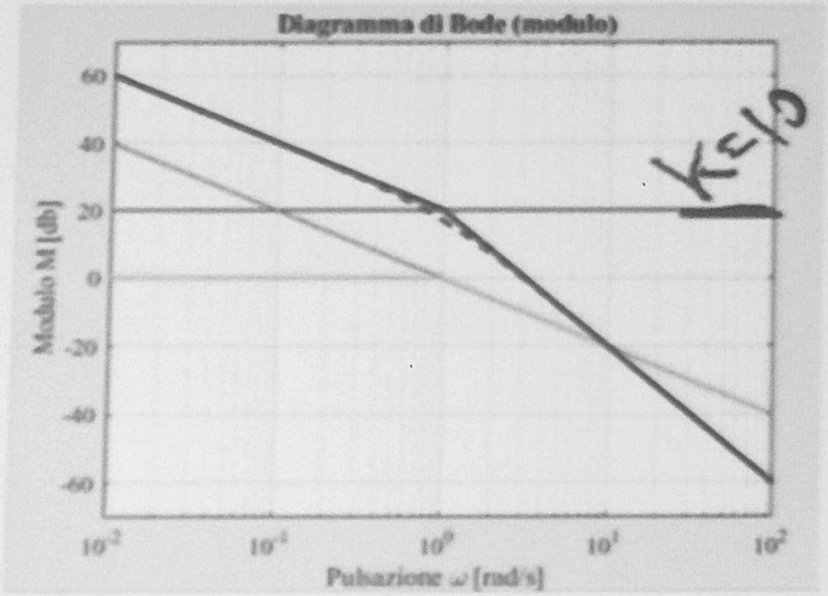



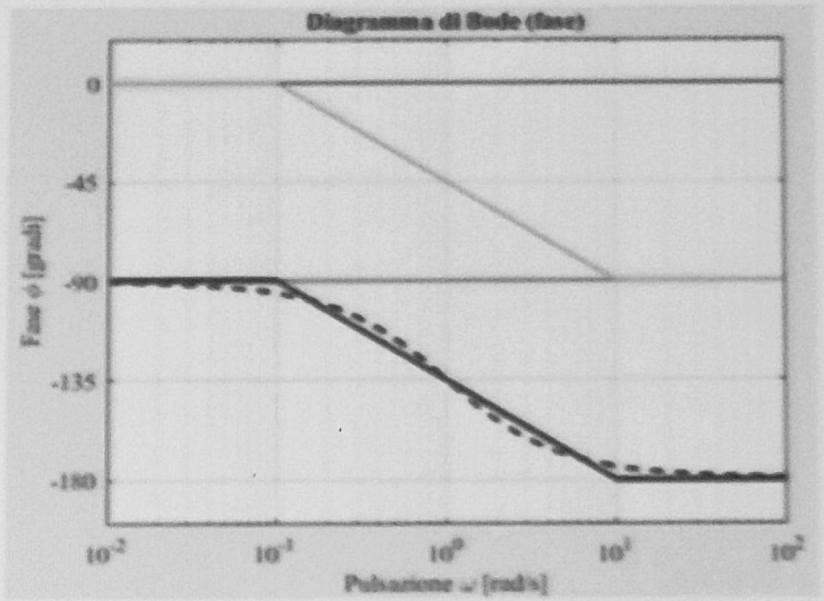



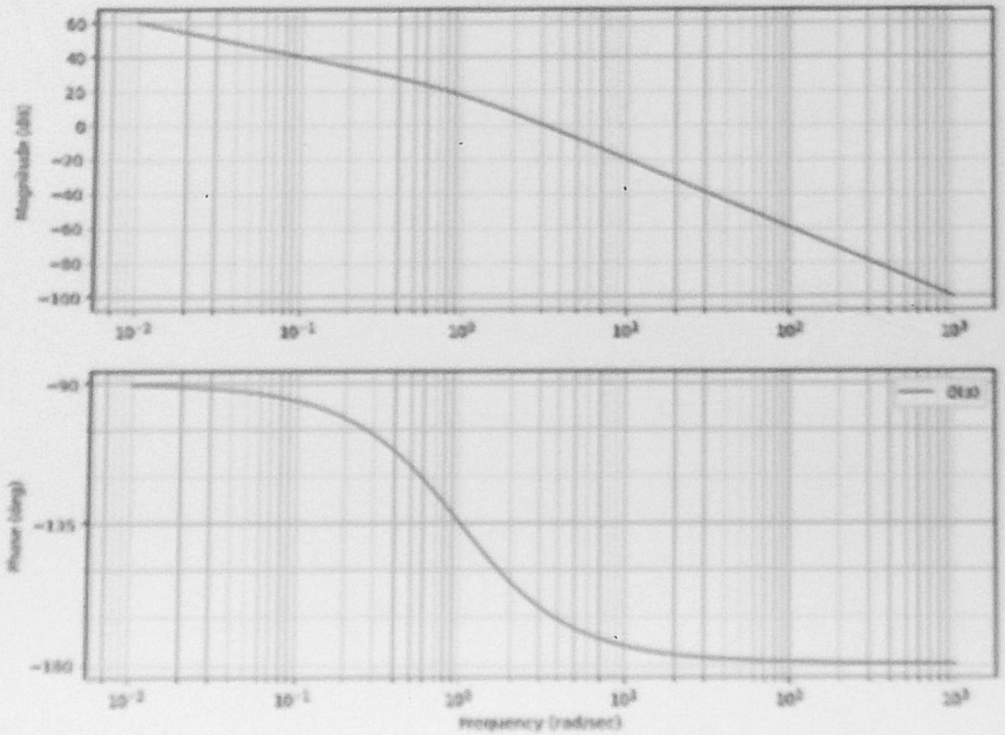

Diagramma di Bode (modulo)
Modulo M [db]
Pulsazione ω [rad/s]


Diagramma di Bode (fase)
Fase φ [gradi]
Pulsazione ω [rad/s]


Magnitude (dB)
Phase (deg)
Frequency (rad/sec)

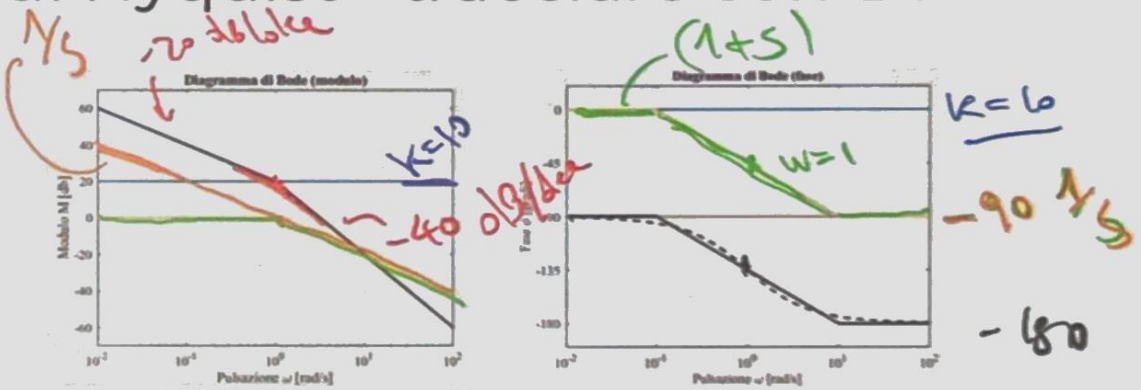

Diagramma di Bode (modulo)
Modulo M [db]
Pulsazione ω [rad/s]
K=1/3

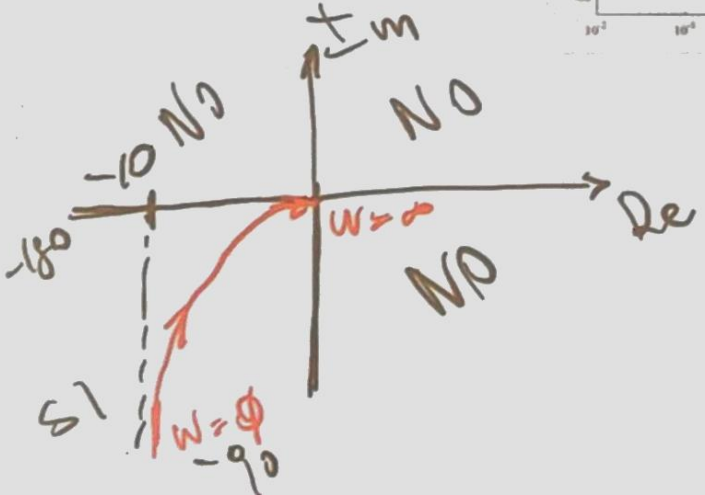

Diagramma di Bode (fase)
Fase φ [gradi]
Pulsazione ω [rad/s]

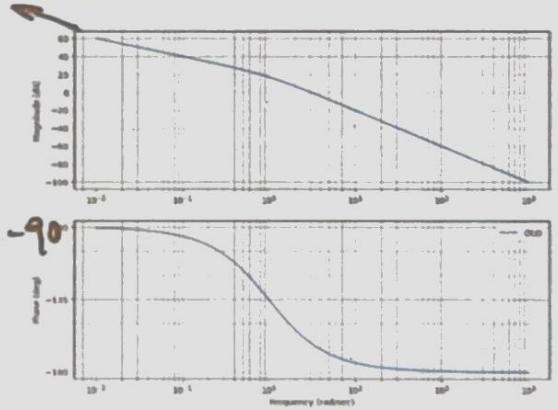

Magnitude (dB)
Phase (deg)
Frequency (rad/sec)

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{10}{s(1+s)}$$



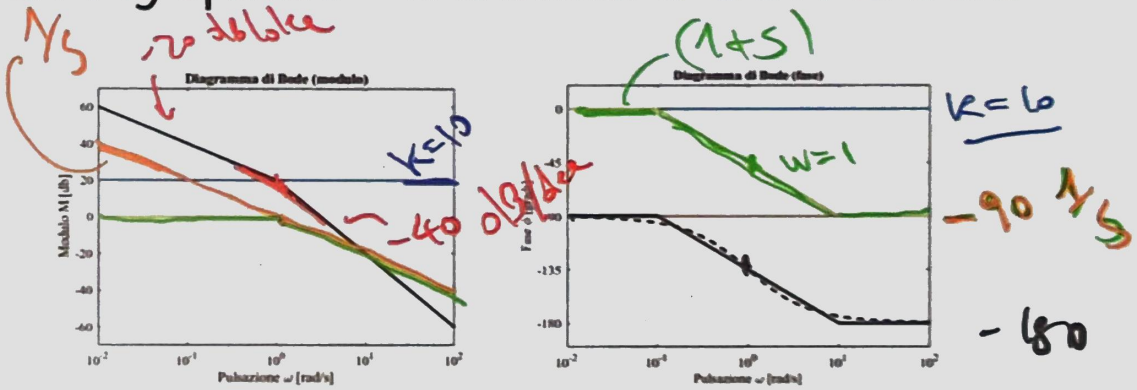



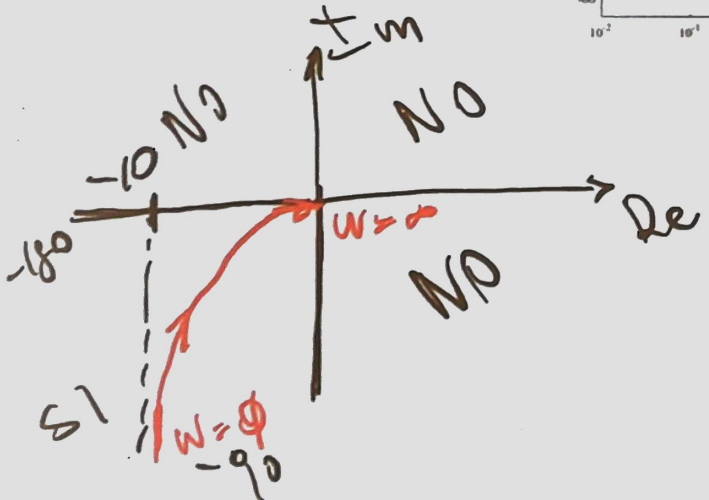



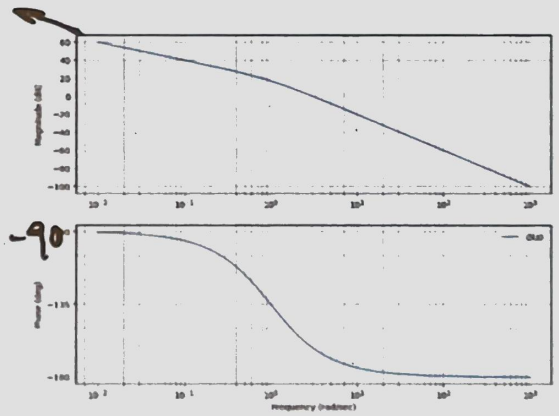

BODE VERO

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{10}{s(1+s)}$$



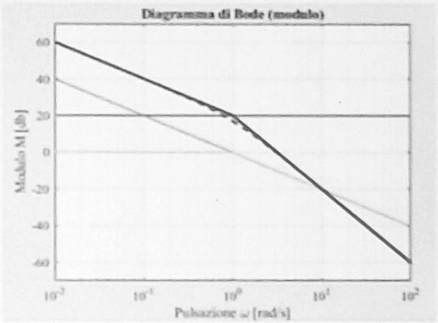



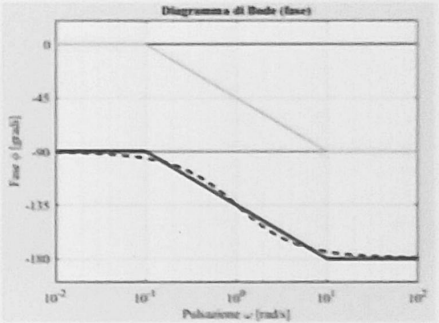



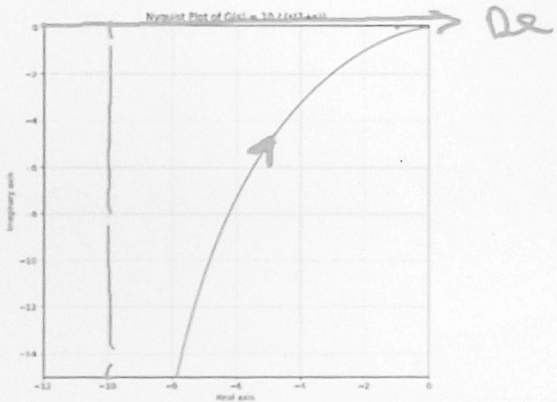

BODE VERO

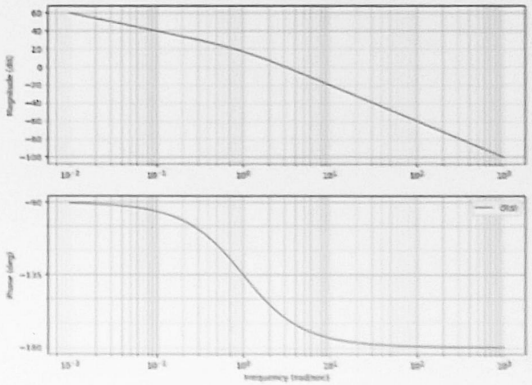
# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{10}{s(1+s)}$$



# I Diagrammi di Nyquist - tracciare con Bode

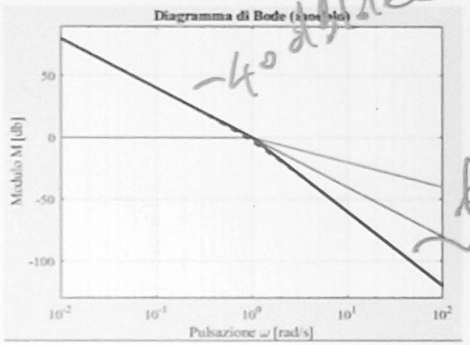
$$G(s) = \frac{1}{s^2(1+s)} \qquad \Rightarrow G(j\omega) = \frac{-1}{\omega^2(1+j\omega)}$$

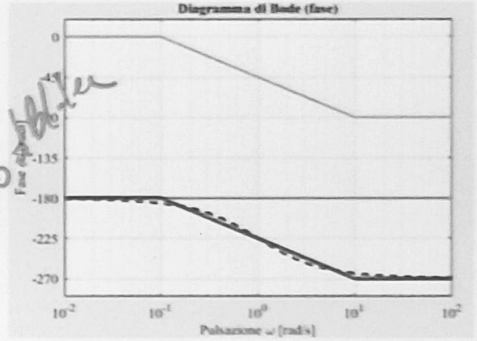
Asintoto

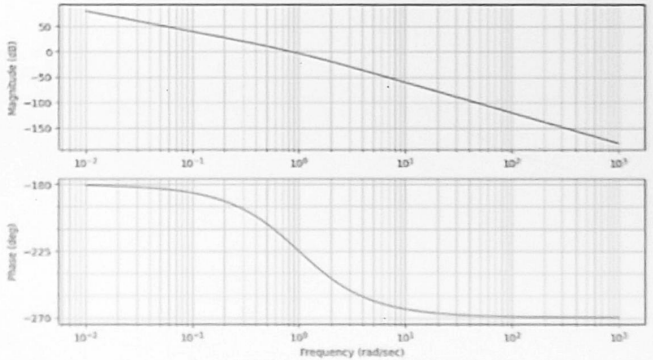
$$\frac{-1}{\omega^2 \cdot (1+j\omega)} \approx_{\omega \to 0} \frac{-(1-j\omega)}{\omega^2} \longrightarrow \begin{cases} \Re\left(\frac{-(1-j\omega)}{\omega^2}\right) = \frac{-1}{\omega^2} \\ \Im\left(\frac{-(1-j\omega)}{\omega^2}\right) = \frac{1}{\omega} \end{cases}$$

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{1}{s^2(1+s)}$$

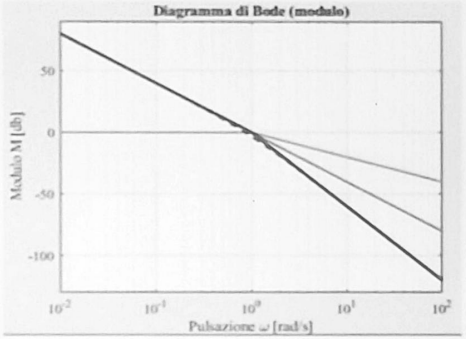


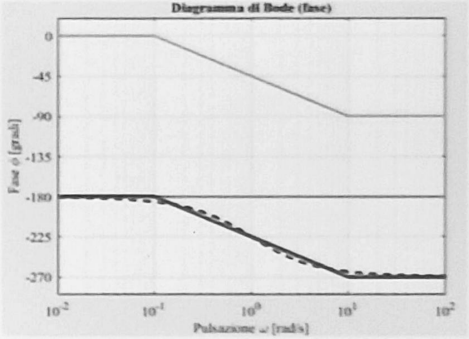



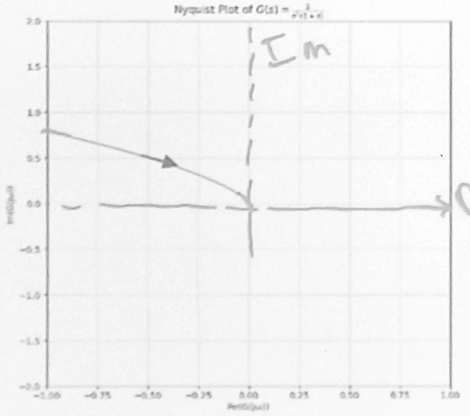



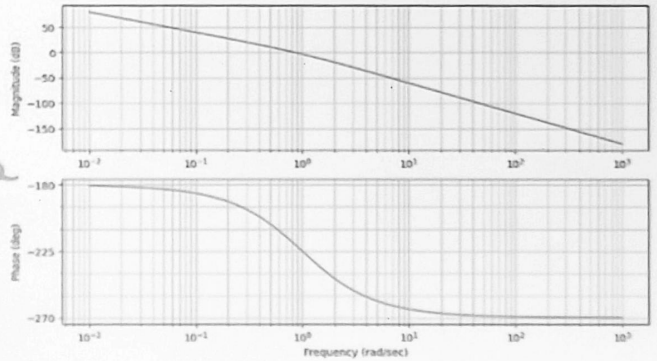
# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{1}{s^2(1+s)}$$



# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{e^{-s}}{s} \quad \Rightarrow \quad G(j\omega) = \frac{e^{-j\omega}}{j\omega} = \frac{\cos(\omega) - j\sin(\omega)}{j\omega}$$

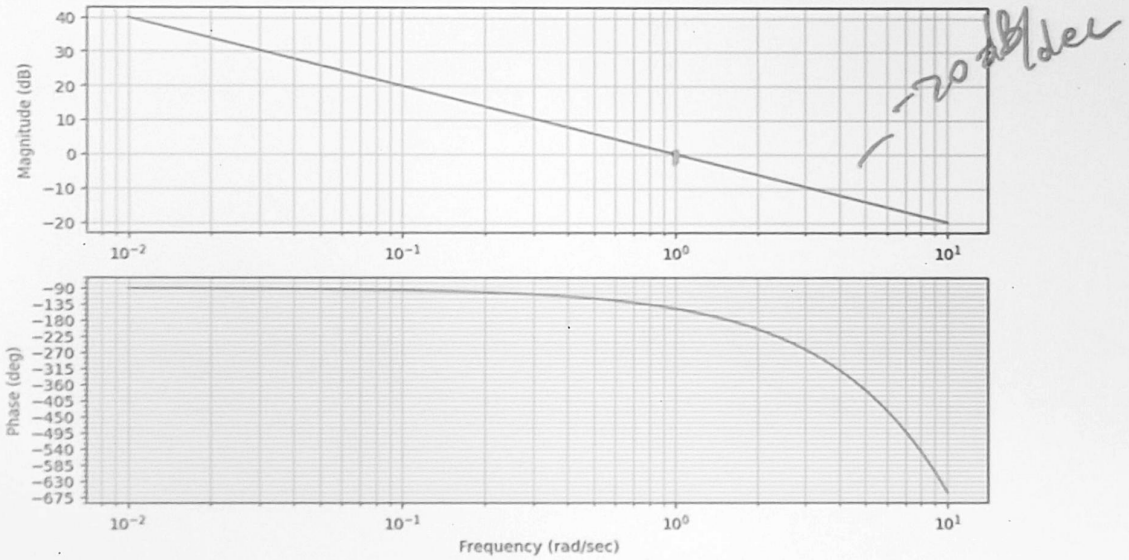
Asintoto $\lim_{\omega \to 0} G(j\omega) = \lim_{\omega \to 0} \frac{-\sin(\omega)}{\omega} = -1$

Guardiamo la fase $\angle G = -\omega - \frac{\pi}{2}$

1. Intersezione asse reale <0: Cerco il valore di w che corrisponde a fase = -180 $\angle G(j\omega) = -\omega - \frac{\pi}{2} = -\pi$
2. Intersezione asse reale <0: Cerco il valore di w che corrisponde a fase = -360

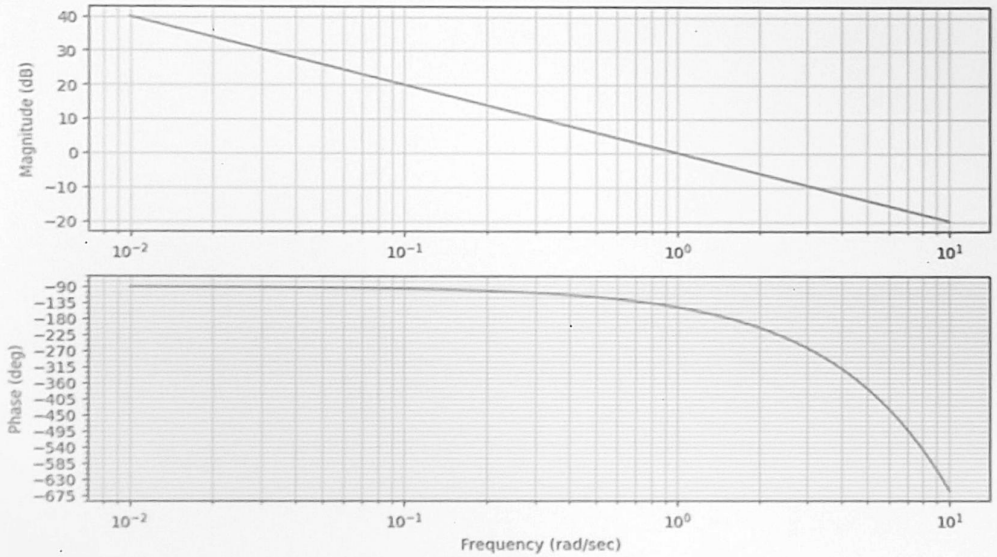
# I Diagrammi di Nyquist - tracciare con Bode

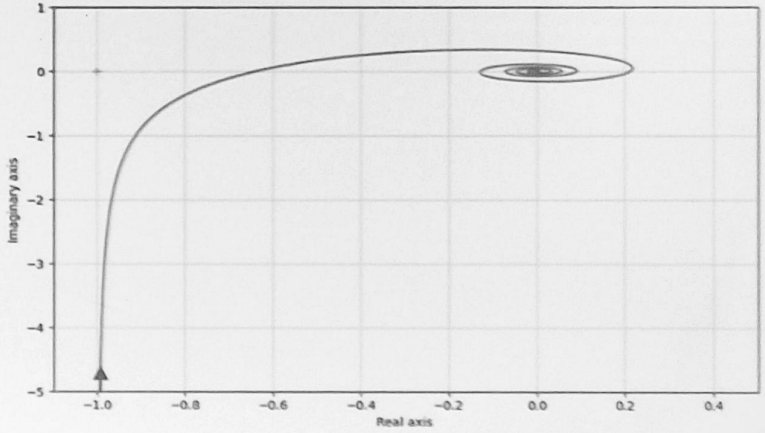
$$G(s) = \frac{e^{-s}}{s} \quad \Rightarrow \quad G(j\omega) = \frac{e^{-j\omega}}{j\omega} = \frac{\cos(\omega) - j\sin(\omega)}{j\omega}$$



# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{e^{-s}}{s} \quad \Rightarrow \quad G(j\omega) = \frac{e^{-j\omega}}{j\omega} = \frac{\cos(\omega) - j\sin(\omega)}{j\omega}$$



# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{(s+10)^2}{s \cdot (s+100)^2} = \frac{1}{100} \cdot \frac{\left(1+\frac{s}{10}\right)^2}{s \cdot \left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100} \cdot \frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega \cdot \left(1+\frac{j\omega}{100}\right)^2}$$

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{(s+10)^2}{s \cdot (s+100)^2} = \frac{1}{100} \cdot \frac{\left(1+\frac{s}{10}\right)^2}{s \cdot \left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100} \cdot \frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega \cdot \left(1+\frac{j\omega}{100}\right)^2}$$

Asintoto

$$\frac{1}{100} \cdot \frac{1}{j\omega} \cdot \frac{\left(1+\frac{j\omega}{10}\right)^2}{\left(1+\frac{j\omega}{100}\right)^2} \approx_{\omega \to 0} \frac{1}{100} \cdot \frac{1}{j\omega} \cdot \left(1+\frac{j\omega}{5}\right) \cdot \left(1-\frac{j\omega}{50}\right)$$

$$1 + \frac{2j\omega}{10} - \frac{\omega^2}{100}$$

$\omega \to 0$

# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{(s+10)^2}{s\cdot(s+100)^2} = \frac{1}{100}\cdot\frac{\left(1+\frac{s}{10}\right)^2}{s\cdot\left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100}\cdot\frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega\cdot\left(1+\frac{j\omega}{100}\right)^2}$$

## Asintoto

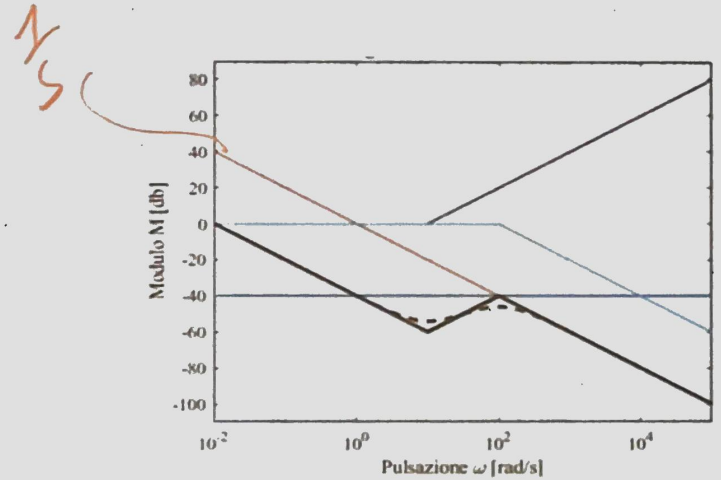
$$\frac{1}{100}\cdot\frac{1}{j\omega}\cdot\frac{\left(1+\frac{j\omega}{10}\right)^2}{\left(1+\frac{j\omega}{100}\right)^2} \approx_{\omega\to 0} \frac{1}{100}\cdot\frac{1}{j\omega}\cdot\left(1+\frac{j\omega}{5}\right)\cdot\left(1-\frac{j\omega}{50}\right) \approx_{\omega\to 0} \frac{1}{100\cdot j\omega}\left(1+\frac{j\omega}{5}-\frac{j\omega}{50}\right) \approx_{\omega\to 0} \frac{1}{100\cdot j\omega}\left(1+\frac{9j\omega}{50}\right)$$

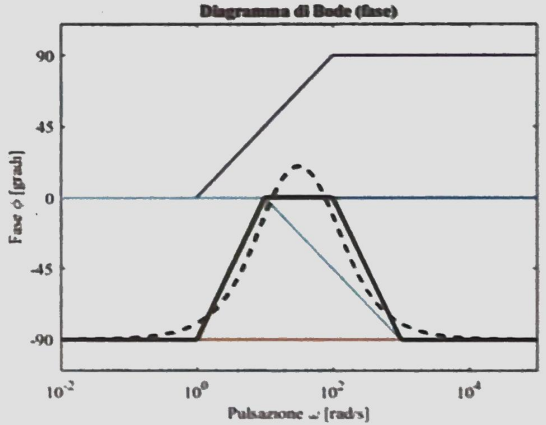
$$1+\frac{2j\omega}{10}-\frac{\omega^2}{100}$$

$$1+\frac{j\omega}{50} \longrightarrow \frac{1}{1+x} \approx 1-x$$

# I Diagrammi di Nyquist - tracciare con Bode

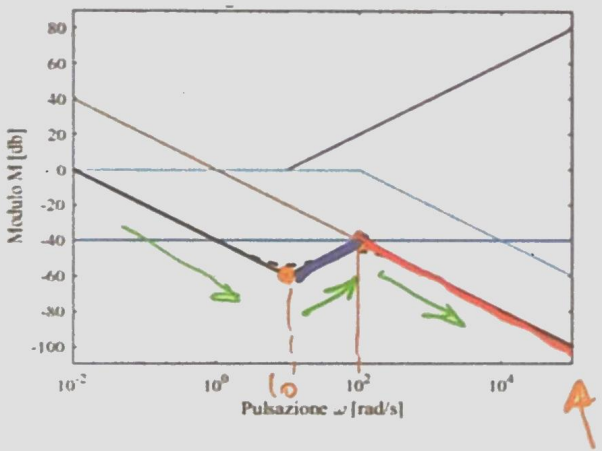
$$G(s) = \frac{(s+10)^2}{s\cdot(s+100)^2} = \frac{1}{100}\cdot\frac{\left(1+\frac{s}{10}\right)^2}{s\cdot\left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100}\cdot\frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega\cdot\left(1+\frac{j\omega}{100}\right)^2}$$

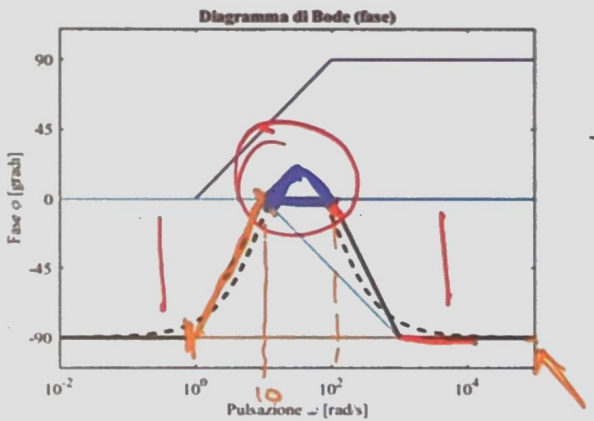


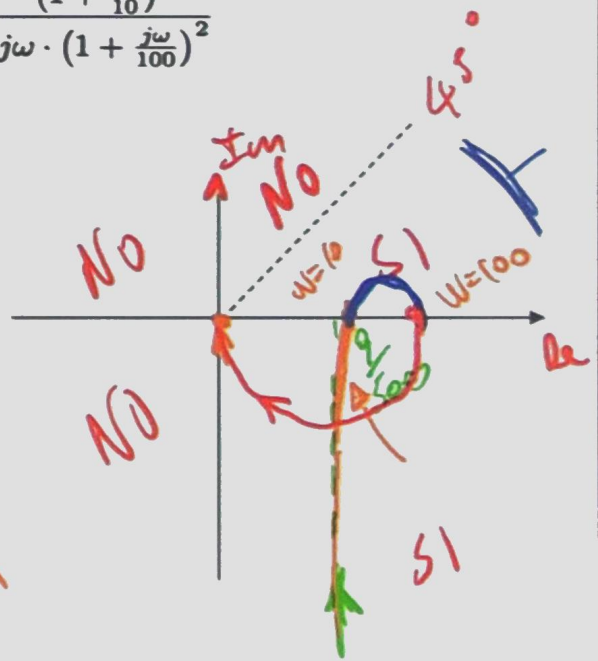



# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{(s+10)^2}{s\cdot(s+100)^2} = \frac{1}{100}\cdot\frac{\left(1+\frac{s}{10}\right)^2}{s\cdot\left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100}\cdot\frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega\cdot\left(1+\frac{j\omega}{100}\right)^2}$$



# I Diagrammi di Nyquist - tracciare con Bode

$$G(s) = \frac{(s+10)^2}{s\cdot(s+100)^2} = \frac{1}{100}\cdot\frac{\left(1+\frac{s}{10}\right)^2}{s\cdot\left(1+\frac{s}{100}\right)^2} \longrightarrow G(j\omega) = \frac{1}{100}\cdot\frac{\left(1+\frac{j\omega}{10}\right)^2}{j\omega\cdot\left(1+\frac{j\omega}{100}\right)^2}$$

